

 Announcement

Kindly note: We're currently experiencing a high volume of inquiries and calls, potentially leading to delays in our initial response.



Dermatology 23S Module 2

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Day 4 follow on Questions

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by [Derek Kayanja \(Tutor\)](#) - Tuesday, 21 November 2023, 2:39 PM

Hi Group,

Please answer at least on of the questions below, if they have all been answered before you have had a chance to respond then comment on your peers posts and ask questions.

1. Discuss clinical features of squamous cell carcinoma and its precursors.
2. Discuss investigations of squamous cell carcinoma.
3. Discuss management of squamous cell carcinoma.

56 words

[Reply](#)**Re: Day 4 follow on Questions**

by [Siddharth Saluja](#) - Wednesday, 22 November 2023, 12:59 PM

1) Clinical Features of Squamous Cell Carcinoma and its Precursors:

Squamous cell carcinoma (SCC) is a type of skin cancer that arises from the squamous cells in the epidermis. It commonly occurs in sun-exposed areas such as the face, lips, ears, and hands. (1) The clinical features of SCC and its precursors, such as actinic keratosis, include:

1. Actinic Keratosis: These are precancerous lesions characterized by rough, scaly patches on the skin. They are often caused by long-term sun exposure and can progress to SCC if left untreated. (2)

2. SCC in situ (Bowen's disease): This is an early form of SCC that is confined to the epidermis. It appears as a

red, scaly patch or plaque that may be crusted or ulcerated. (2)

3. Invasive SCC: As SCC progresses, it can invade deeper layers of the skin and surrounding tissues. It typically presents as a firm, raised nodule or ulcer with a rough surface. It may bleed, fail to heal, or develop a crust. (2)

2) Investigations of Squamous Cell Carcinoma:

To diagnose and stage SCC, the following investigations may be performed: (3)

1. Biopsy: A skin biopsy is essential to confirm the diagnosis of SCC. A sample of the suspicious lesion is taken and examined under a microscope to determine the presence of cancerous cells. (3)

2. Imaging: Imaging studies like ultrasound, CT scan, or MRI may be ordered to assess the extent of tumour invasion, involvement of nearby structures, and the presence of regional lymph node metastasis. (3)

3. Sentinel Lymph Node Biopsy: In cases where there is a high risk of lymph node involvement, a sentinel lymph node biopsy may be performed to determine if cancer has spread to the nearby lymph nodes. (3)

3) Management of Squamous Cell Carcinoma:

The management of SCC depends on various factors, including the tumor size, location, depth of invasion, involvement of lymph nodes, and overall health of the patient. Treatment options may include 🧐 4)

1. Surgical Excision: The primary treatment for SCC is surgical excision. This involves removing the tumor along with a margin of healthy tissue to ensure complete removal. Mohs micrographic surgery may be used for tumors in cosmetically sensitive areas.(4)

2. Radiation Therapy: Radiation therapy may be recommended for SCC that is large, invasive, or has a high risk of recurrence. It can be used as the primary treatment or in combination with surgery. (4)

3. Chemotherapy: In advanced cases or when SCC has spread to distant organs, chemotherapy drugs may be used to help control the disease and alleviate symptoms. (4)

4. Immunotherapy: In some cases, immunotherapy drugs may be used to stimulate the body's immune system to recognize and attack cancer cells. (4)

The choice of treatment depends on individual patient factors and should be discussed with a healthcare professional who can provide personalized recommendations based on the specific situation. Regular follow-up and surveillance are important to monitor for recurrence or metastasis.

References:

- 1) Cleveland Clinic. (2022). Squamous Cell Carcinoma: Symptoms, Causes & Treatment. [online] Available at: <https://my.clevelandclinic.org/health/diseases/17480-squamous-cell-carcinoma>.
- 2) The James - OSUCCC. (n.d.). Squamous Cell Carcinoma Testing & Diagnosis | OSUCCC – James. [online] Available at: <https://cancer.osu.edu/for-patients-and-caregivers/learn-about-cancers-and-treatments/cancers-conditions-and-treatment/cancer-types/skin-cancers/squamous-cell-carcinoma/screening-and-diagnosis>.
- 3) Combalia, A. and Carrera, C. (2020). Squamous Cell Carcinoma: An Update on Diagnosis and Treatment. Dermatology Practical & Conceptual, [online] 10(3). doi:<https://doi.org/10.5826/dpc.1003a66>.
- 4) Firnhaber, J.M. (2012). Diagnosis and Treatment of Basal Cell and Squamous Cell Carcinoma. American Family Physician, [online] 86(2), pp.161–168. Available at: <https://www.aafp.org/pubs/afp/issues/2012/0715/p161.html>.

569 words

Reply



Re: Day 4 follow on Questions

by [Jillian Simpson](#) - Wednesday, 22 November 2023, 5:29 PM

Thanks Sid.

I wanted to expand on the clinical features of squamous cell carcinomas (SCC) some more. SCC is the second commonest skin cancer after basal cell carcinoma (BCC) here in the UK (Morris-Jones, 2019).

70% of lesion occur on the head and neck (Morris-Jones, 2019). They usually present in elderly patients on sun exposed sites and are three times commoner in men (Burge, Wallis, Martin, 2016). Clinical suspicion of an SCC arises when lesions are rapidly growing-they typically grow over weeks to months (Oakley, 2015). They also tend to be painful and hyperkeratotic (Morris-Jones, 2019). SCCs are usually nodular with surface changes including crusting, ulceration, or the formation of a cutaneous horn.

Some lesions can be verrucous and therefore mistaken for viral warts, there is also potential for them to arise in a chronic viral wart (Morris-Jones, 2019). I have included some visuals to highlight the variety of presentations you might see with these lesions.

All patients should have their regional lymph nodes palpated to look for local metastases for any lesions suspicious of an SCC (Morris-Jones, 2019).

Actinic keratosis is the most common precursor lesion for SCC (Oakley, 2015). Again these are usually found on sun-exposed sites, mainly face, ears and hands. Although all the ones I have seen in practice have been on bald scalps. Clinically, their appearance varies from a rough area of skin to a raised keratotic lesion (Morris-Jones, 2019). They usually have an irregular edge and are often less than 1 cm in diameter. They are often multiple and have a hard scaly surface without induration. The degree of scale and redness is variable, and the lesions frequently remit spontaneously (Oakley, 2015).

My question for the group:

- 1. All the images I have found of these lesions have be in Caucasian skin types. Has anyone seen an SCC in skin of colour and can you describe the features or share any images?**
- 2. Does a SCC which presents as a cutaneous horn generally carry a better prognosis than one which is ulcerated?**

References:

Morris-Jones, R. (2019). ABC of Dermatology. Seventh Edition. Wiley Blackwell. P198

Burge, S.M., Wallis, D. and Matin, R. (2016). Oxford handbook of medical dermatology. Oxford: Oxford University Press. p350-351.

Oakley, A. (2015). Cutaneous squamous cell carcinoma | DermNet NZ. [online] Available at: <https://dermnetnz.org/topics/cutaneous-squamous-cell-carcinoma>. [Accessed 22/11/23].

Oakley, A. (2015). Common skin lesions. Squamous cell carcinoma | DermNet. [online] Available at: <https://dermnetnz.org/cme/lesions/squamous-cell-carcinoma>. [Accessed 22/11/23].

401 words





Reply



Re: Day 4 follow on Questions

by [Sadaf Basharat](#) - Thursday, 23 November 2023, 8:03 PM

Hi Jilian, thanks for asking an interesting question.
I have tried to answer your second question as follow.

Does a SCC which presents as a cutaneous horn generally carry a better prognosis than one which is ulcerated?

It's crucial to consider histopathological analysis for a valid prognosis in cases of cutaneous horns, given the potential for underlying malignant or premalignant conditions. The association between actinic keratoses (AK) and squamous cell carcinoma (SCC) suggests they could be common primary disease processes with carcinogenic potential. Yu et al

References:

1. Iro, S., Maadane, A. and Slimani, F., 2021. Facial cutaneous horn in three different conditions: A highlight of a malignancy risk. *International Journal of Surgery Case Reports*, 83, p.105999.

117 words

[Reply](#)



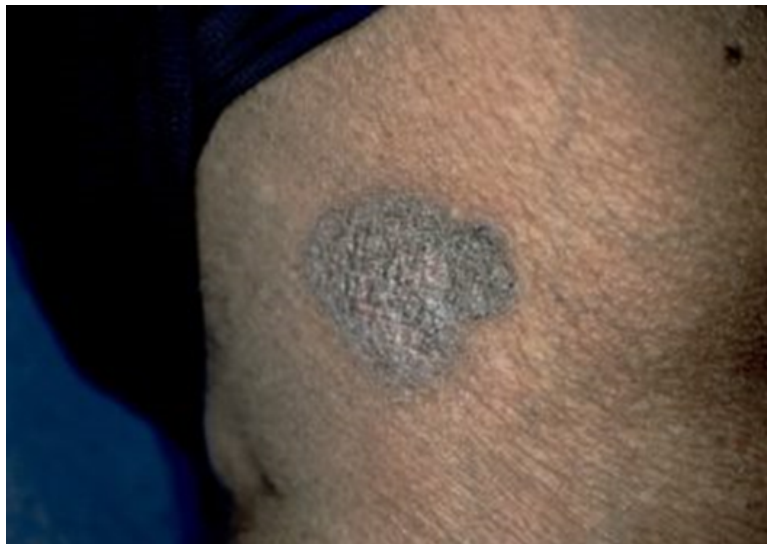
Re: Day 4 follow on Questions

by [Pooran Outar](#) - Friday, 24 November 2023, 1:56 AM

Hello Jillian,

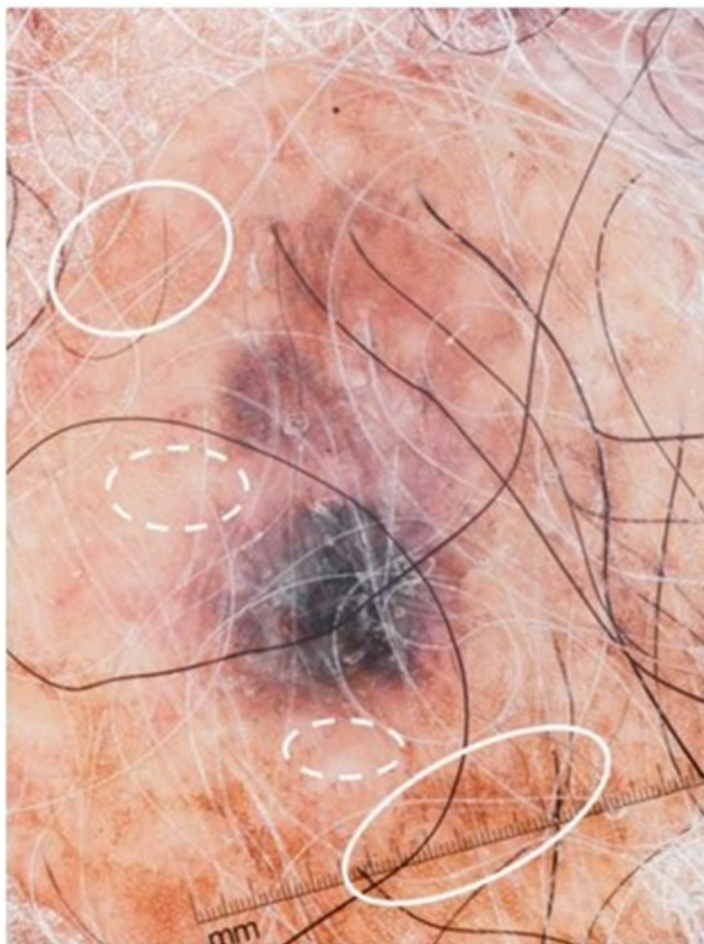
Your post highlights the significance of the clinical features of SCC and as the second most prevalent skin cancer in the UK. While specific data on SCC in Guyana is not readily available, it also appears to be less common than Melanoma. Given that a majority of our population has a diverse skin tone, your inquiry about SCC in individuals with coloured skin is particularly relevant to my clinical practice.

Studies indicate that SCC is relatively rare in individuals of African or Asian descent. However, it's noteworthy that SCC in Black individuals carries a higher mortality rate, potentially attributed to delayed diagnosis. This delay may be due to tumours occurring in sun-protected areas, as indicated by Wells, 2022. Additionally, in individuals with Black and brown skin, squamous cell carcinomas are more likely to manifest on areas of the skin not exposed to the sun, such as the genitals (Mayo Clinic, 2023).



Squamous Cell Carcinoma in an individual of colour (Skin Cancer Foundation, 2022)

Regarding the characteristics of SCC in skin of colour, Mancini et al., 2022 found "that among the dermoscopic images of 8 SCCs, a significant number exhibited intense pigmentation. In the adjacent unaffected skin, 50% displayed a pigmented reticular network accompanied by a 'halo' effect, while the remaining half exhibited sporadic hyperpigmentation."



Dermascopy of SCC in an individual of coloured skin (Manci et al., 2022)

References:

- Manci, R., Dauscher, M., Marchetti, M.A., Usatine, R., Rotemberg, V., Dusza, S. and Marghoob, A. (2022). Features of Skin Cancer in Black Individuals: A Single-Institution Retrospective Cohort Study. *Dermatology Practical & Conceptual*, p.e2022075. doi:<https://doi.org/10.5826/dpc.1202a75>.
- Mayo Clinic (2023). *Squamous cell carcinoma of the skin - Symptoms and causes*. [online] Mayo Clinic. Available at: <https://www.mayoclinic.org/diseases-conditions/squamous-cell-carcinoma/symptoms-causes/syc-20352480>.
- Skin Cancer Foundation (2022). *Skin Cancer & Skin of Color*. [online] The Skin Cancer Foundation. Available at: <https://www.skincancer.org/skin-cancer-information/skin-cancer-skin-of-color/>.
- Wells, J.W. (2022). Cutaneous Squamous Cell Carcinoma: Practice Essentials, Background, Pathophysiology. *eMedicine*. [online] Available at: https://emedicine.medscape.com/article/1965430-overview?form=fpf&scode=msp&st=fpf&socialSite=google&icd=login_success_gg_match_fpf#a5 [Accessed 24 Nov. 2023].

321 words

Reply



Re: Day 4 follow on Questions

by [Jillian Simpson](#) - Friday, 24 November 2023, 10:53 AM

Hi Sadaf and Pooran

Thank you both for taking the time to respond to my questions and providing further detail, along with the images too.

Jill

26 words

[Reply](#)**Re: Day 4 follow on Questions**by [Derek Kyanja \(Tutor\)](#) - Friday, 24 November 2023, 12:29 PM

Pooran, thank you for your response to Jillians question.

9 words

[Reply](#)**Re: Day 4 follow on Questions**by [Derek Kyanja \(Tutor\)](#) - Friday, 24 November 2023, 12:28 PM

Thank you Jillian, good post, good pictures and good follow on question. Please read this articles
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2757062/>
and

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9060139/>

19 words

[Reply](#)**Re: Day 4 follow on Questions**by [Derek Kyanja \(Tutor\)](#) - Friday, 24 November 2023, 12:12 PM

Thank you Siddharth, very good post (but please try and use Harvard referencing).

13 words

[Reply](#)**Re: Day 4 follow on Questions**by [Siddharth Saluja](#) - Friday, 24 November 2023, 4:39 PM

sure Derek

2 words

[Reply](#)**Re: Day 4 follow on Questions**by [Felica Mwinji Namukonde](#) - Sunday, 26 November 2023, 4:09 PM

i wanted to add on management of squamous cell carcinoma

Management Approach:

Risk assessment;

Once you establish diagnosis of cSSC by biopsy of the skin and histopathology doing risk assessment of locoregional recurrence or distant metastasis is very important step to determine treatment.

Initial evaluation of patients must include physical examination of local and regional lymph nodes, mostly if this examination is negative no further work up is required.

Risk of recurrence characteristics include;

Tumor site, depth, size and histological features, comorbidities and patient charectaristics.

Categorizing tumors as low risk or high risk determine treatment approach. The goals of treatment of primary cSCC is make sure the primary tumor is removed, prevent metastasis and preserve cosmesis especially for tumors in areas located in functional challenging sites.

Single Low-Risk Lesions;

Surgery

Standard excision this is first-line treatment for low-risk cSCCs, National Comprehensive Cancer Network (NCCN) guidelines and American Academy of Dermatology recommends 4 to 6mm clinical margins of uninvolved skin for standard excision and depth of mid-deep subcutaneous adipose tissue. For incompletely excised cSCC circumferential histologic margin or Mohs must be advised, in surgery contraindicated patients consider adjuvant radiation therapy.

Mohs Surgery; a specialized tissue sparing procedure involves histological assessment of 100% of the excised tumor margins. May also be used to treat tumors in locations which need tissue-sparing surgical morbidity and cosmetically sensitive areas.

Nonsurgical destructive treatments;

Curettage and Electrodesiccation; best in small, well-defined cSCCs in non critical low-risk sites. It has low complication rate, not expensive and favorable cosmetic results.

Disadvantage is lack of confirming tumor margins, which then limits its use to small well defined nonrecurrent low risk tumors.

CRYOTHERAPY; destroys malignant cells by freezing and thawing. Tumor cell death happens by forming intracellular and extracellular ice crystals.. hypertonicity, disruption of phospholipid membrane and vascular stasis. Cryotherapy does not allow confirmation of adequacy of margins to be treated it is used only in small well-defined and low risk invasive cSCCs and Bowens disease.

Photodynamic therapy; porphyrin produce cytotoxicity in presence of oxygen after light stimulation of appropriate wavelength. Not recommended in treating invasive cSCC due to high recurrence rates.

Radiation therapy not used as monotherapy for low risk tumors indicated in primary cSCC in certain patients.(Aasi S Z)

Group Question can someone outline treatment approach for High-Risk and very High-Risk cSCCs?

References

DeSimone J A Hong A M (2023) *Uptodate* Available at: https://www.uptodate.com/contents/recognition-and-management-of-high-risk-aggressive-cutaneous-squamous-cell-carcinoma?search=Recognition%20and%20management%20of%20high%20risk%20cutaneous%20squamous%20cell%20carcinoma&source=search_result&selectedTitle=1~150&usage Accessed (26/11/23)

Aasi S Z, Hong A M (2023) *Uptodate* Available at: https://www.uptodate.com/contents/treatment-and-prognosis-of-low-risk-cutaneous-squamous-cell-carcinoma-csc?search=treatment%20and%20prognosis%20low-risk%20cutaneous%20squamous%20cell%20carcinoma&source=search_result&selectedTitle=1~150&usage Accessed (26/11/23)

411 words

Reply



Re: Day 4 follow on Questions

by [Sadaf Basharat](#) - Thursday, 23 November 2023, 7:38 PM

1. Discuss investigations of squamous cell carcinoma.

The diagnosis of SCC starts from getting a detailed history and examination of the lesion. SCC is usually found on non sun exposed area and history should be focused on any underlying Further investigations to diagnose SCC includes :

1. Dermoscopy: Dermoscopy is usually done as a first line investigation for the diagnosis of SCC.

2. Excision and biopsy of the lesion: The confirmed diagnosis of SCC is made by excision and biopsy of the lesion. Biopsies that extend into the mid-reticular dermis are usually preferred for the evaluation of invasive disease.(Lim and Asgari, 2022)

3. For further management of the SCC, staging investigations are required to ascertain the extent of the disease and its probable prognosis. These may include imaging, using USG, Xrays, CT scans or MRI scans. Biopsies of lymph nodes and other tissue are also included. (Oakley, 2015)

Question for the group. Could anyone elaborate the diagnostic findings of Dermoscopy for SCC?

References:

1. Oakley, A. (2015). *Cutaneous squamous cell carcinoma* / DermNet NZ. [online] dermnetnz.org. Available at: <https://dermnetnz.org/topics/cutaneous-squamous-cell-carcinoma>
2. Lim, J. and Asgari, M. (2022) Cutaneous squamous cell carcinoma UpToDate. Available at: https://www.uptodate.com/contents/cutaneous-squamous-cell-carcinoma-cscc-clinical-features-and-diagnosis?search=cutaneous%20squamous%20cell%20carcinoma%20investigations&source=search_result&selectedTitle

188 words

Reply



Re: Day 4 follow on Questions

by [Derek Kayanja \(Tutor\)](#) - Friday, 24 November 2023, 12:28 PM

Sadaf, Thank you for your input to the discussion.

9 words

Reply



Re: Day 4 follow on Questions

by [Arwaf Benrawwaf](#) - Friday, 24 November 2023, 4:32 PM

Hi Sadaf, Thank you for your post.

Dermoscopy can aid in the diagnosis of SCC by revealing specific features that are suggestive of the lesion being malignant. These features include: (Masson, E. 2018) (Yélamos, O. 2019)

1. Keratin pearls: These are white or yellowish round structures that represent areas of keratinization within the tumor.
2. Ulceration: Ulceration is a common feature of SCC and is seen as a central area of erosion or crusting within the lesion.
3. Telangiectasia: Telangiectasia refers to the presence of dilated blood vessels within or around the lesion. It is a common finding in SCC.
4. Atypical vessels: These are vessels that are irregular in shape, size, or distribution. They may be indicative of malignancy.
5. Irregular pigmentation: SCC may exhibit irregular pigmentation, which can be seen as areas of brown, black, or gray color within the lesion.
6. Asymmetry: SCC lesions may be asymmetric in shape, with irregular borders and varying colors.

Dermoscopy can also help differentiate SCC from other skin lesions such as seborrheic keratosis, actinic keratosis, and basal cell carcinoma. However, a definitive diagnosis of SCC still requires histopathologic examination of a biopsy specimen. (Yélamos, O. 2019)

References:

- Masson, E. (n.d.). *Guidelines of care for the management of cutaneous squamous cell carcinoma*. [online] EM-Consulte. Available at: <https://www.em-consulte.com/article/1200008/guidelines-of-care-for-the-management-of-cutaneous> [Accessed 24 Nov. 2023].

- Yélamos, O., Braun, R.P., Liopyris, K., Wolner, Z.J., Kerl, K., Gerami, P. and Marghoob, A.A. (2019). Dermoscopy and dermatopathology correlates of cutaneous neoplasms. *Journal of the American Academy of Dermatology*, 80(2), pp.341–363. doi:<https://doi.org/10.1016/j.jaad.2018.07.073>.

253 words

Reply



Re: Day 4 follow on Questions

by [Sophie Holloran](#) - Saturday, 25 November 2023, 11:09 AM

Hi everyone,

I thought I would add a bit more about cSCC in situ – **Bowens disease**

Bowens disease presents as a well demarcated red patch or plaque, although can be skin coloured in people with darker skin types. They occur in sun exposed sites, such as the ears, face, hands and lower legs. They can appear like small patches of inflammatory disease, hence can be mistaken for eczema or psoriasis.

Dermoscopy of Bowens disease:

- Dotted and glomerular vessels
- White or yellow surface scale
- Brown pattern with hypo-pigmented areas

Histology of bowens disease:

- Keratinocytic dysplasia
- Thickening of the epidermis = acanthosis

Treatment of bowens disease:

It has been reported that the incidence of invasive SCC is 5% in people with bowens disease.

The management options include:

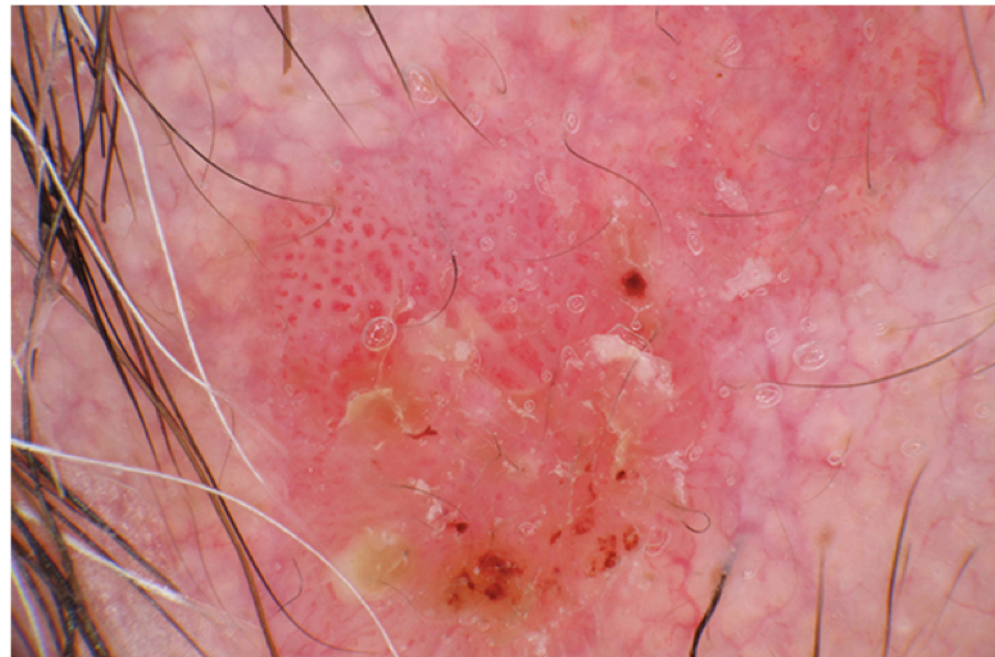
1. Watch and wait – this may be a good treatment strategy for elderly or frail patients that wouldn't be able to tolerate other treatments, given that the risk of SCC development is very low.
2. Surgical excision
3. Shave excision
4. Curettage and cautery
5. Cryotherapy
6. Efudix topical treatment
7. Imiquimod topical treatment
8. Photodynamic therapy

Bowen's disease (cutaneous squamous cell carcinoma in situ)

An erythematous, scaly, thin plaque is present on the skin.

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Graphic 82358 Version 7.0

Dermoscopic image of nonpigmented squamous cell carcinoma in situ (Bowen disease)

Large, coiled (glomerular) vessels in a grouped arrangement and yellow surface scales are the dermoscopic hallmarks of nonpigmented Bowen disease.

Graphic 96298 Version 2.0

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Questions for the group:

**Does anyone have any practical tips for recognising Bowens disease on dermoscopy?
Given there are so many management choices, what options do you choose and why?**

References:

1. Coulson (2023) Intraepidermal squamous cell carcinoma - Derm Net NZ. Available at: <https://dermnetnz.org/topics/intraepidermal-squamous-cell-carcinoma> (Accessed 25/11/23)
2. Lim (2023) Cutaneous squamous cell carcinoma (cSCC): clinical features and diagnosis - Up To Date. Available at: https://www.uptodate.com/contents/cutaneous-squamous-cell-carcinoma-cscc-clinical-features-and-diagnosis?search=bowen%20disease&source=search_result&selectedTitle=1~61&usage_type=default&display_rank=1#H (Accessed 25/11/23)

250 words

Reply

**Re: Day 4 follow on Questions**

by [Derek Kayanja \(Tutor\)](#) - Saturday, 25 November 2023, 3:19 PM

Thank you Sophie, this is a very useful addition to our discussion.

12 words

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